A Mini Project Report on

"RHYTHMIX"

Submitted in partial fulfillment of the requirement for the Sixth Semester

Bachelor of Engineering
In
Computer Science and Engineering
Visvesvaraya Technological University, Belgaum

Submitted by

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CERTIFICATE

This is to certify that the mini project entitled "RHYTHMIX" is a bonafide work carried out by "MANOJ M,VIGNESH N, SUHAAS REDDY" in partial fulfilment of 6th semester, **Bachelor of Engineering** in **Computer Science and Engineering** under **Visvesvaraya Technological University, Belgaum** during the year 2022-23.

Dr. Nagaraja J **Associate Professor** **Dept. of CSE, DSCE** **Bangalore** **Signature: _____**	**Dr. Ramesh Babu D R** **Vice Principal & HOD,** **Dept of CSE, DSCE** **Bangalore.** **Signature: _____**	**Dr. B G Prasad** **Principal,DSCE,** **Bangalore.** **Signature: _____**

Name of the Examiners	**Signature with date**
1…….	……...
2…….	……...

ACKNOWLEDGEMENT

We are pleased to have successfully completed the Mini project **"RHYTHMIX"**. We thoroughly enjoyed the process of working on this project and gained a lot of knowledge doing so.

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Minor Project- Report
March 2023-July 2023

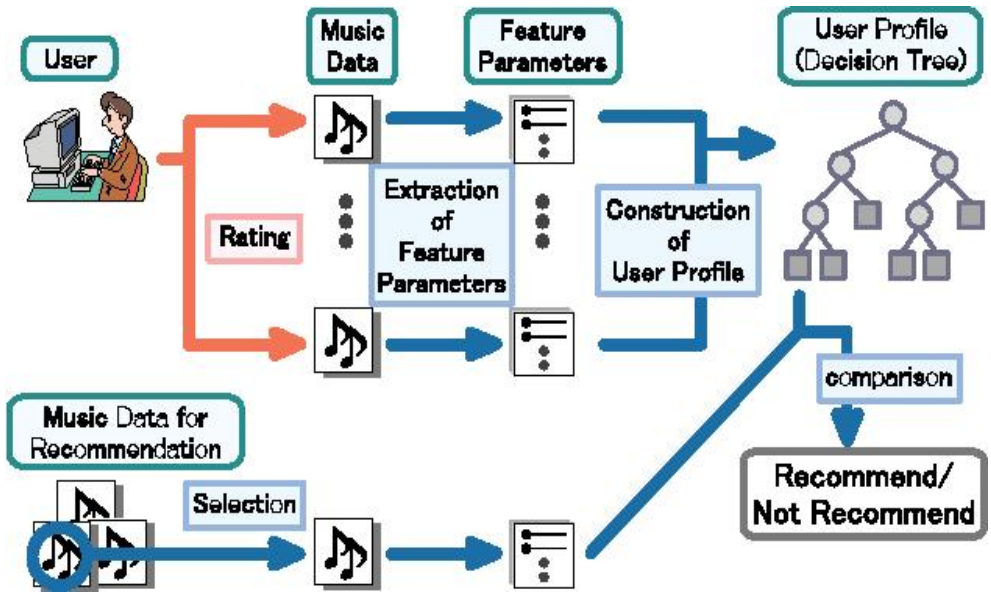
Course Faculty: DR. NAGARAJA J

Course Name & code: Mini Project, 19CS6DCMIP

Semester: VI

Date: 16/06/2023

TITLE OF THE PROJECT	RHYTHMIX- Music Recommendation System			
STUDENT NAME	MANOJ M	VIGNESH N	SUHAAS REDDY	
USN	1DS20CS119	1DS20CS130	1DS20CS159	
INDIVIDUAL CONTRIBUTION	CONTENT BASED FILTERING	POPULARITY BASED FILTERING	GATHERING AND LOADING DATASETS	
GUIDE	DR. NAGARAJA J			
PROJECT ABSTRACT :	Rhythmix is a music recommendation system that revolutionizes the way users discover new songs. By leveraging both user data and extensive song information, Rhythmix offers personalized recommendations based on previous user and song information. Through popularity-based and content-based filtering algorithms, the system ensures accurate and diverse song suggestions. By analyzing historical listening patterns and comprehensive song data, Rhythmix identifies songs with similar attributes and considers popular trends, providing users with an engaging and satisfying music discovery experience. With its advanced recommendation algorithm, Rhythmix adapts to evolving preferences, connecting users to the vast world of music in a more meaningful way.			
PLATFORM USED (H/W & S/W TOOLS USED	Pandas, Recommenders, Jupyter Notebook, VS Code, MS Excel.			

INTRODUCTION	Rhythmix, a state-of-the-art music recommendation system designed to enhance the music discovery experience for users. Rhythmix leverages the power of algorithms and data analysis to curate personalized song recommendations that align with each user's unique musical inclinations. By harnessing the wealth of information derived from both user data and comprehensive song attributes, Rhythmix offers a seamless and engaging platform for users to explore new music.With Rhythmix, users can bid farewell to generic playlists and dive into a world of customized musical journeys. The system takes into account each user's historical listening patterns, preferences, and feedback, allowing it to understand their individual taste profiles.
LIST OF PAPERS/URLS REFERRED	1. Content-based filtering for music recommendation: A comprehensive survey by Bogdanov, D., Wack N., et al. (2013). 2. A survey on content-based filtering methods for efficient recommendation systems by Gupta, M., Mittal, N. (2016). 3. Content-based filtering using machine learning techniques for effective recommender systems" by Cambazoglu, B.B., et al. (2010).
DESIGN {SYSTEM DESIGN DIAGRAM}	
PROJECT SOURCE CODE LINK (GITHUB/ GOOGLE DRIVE)	https://github.com/Vignesh6236/Rhythmix

CONCLUSION /FUTURE ENHANCEMENT	Rhythmix is a recommendation system that recommends songs based on user data and song data using popularity based filtering and content based filtering.Future improvements would be to employ additional algorithms like collaborative filtering techniques.
UI SCREENSHOTS OF SAMPLE RESULTS	

```
In [49]: pr.recommend(song_df['user_id'][5])
```

Out[49]:

	user_id	song	score	Rank
3660	b80344d063b5ccb3212f76538f3d9e43d87dca9e	Sehr kosmisch - Harmonia	45	1.0
4678	b80344d063b5ccb3212f76538f3d9e43d87dca9e	Undo - Bj&#65533;rk	32	2.0
5105	b80344d063b5ccb3212f76538f3d9e43d87dca9e	You're The One - Dwight Yoakam	32	3.0
1071	b80344d063b5ccb3212f76538f3d9e43d87dca9e	Dog Days Are Over (Radio Edit) - Florence + Th...	28	4.0
3655	b80344d063b5ccb3212f76538f3d9e43d87dca9e	Secrets - OneRepublic	28	5.0
4379	b80344d063b5ccb3212f76538f3d9e43d87dca9e	The Scientist - Coldplay	27	6.0
4712	b80344d063b5ccb3212f76538f3d9e43d87dca9e	Use Somebody - Kings Of Leon	27	7.0
3476	b80344d063b5ccb3212f76538f3d9e43d87dca9e	Revelry - Kings Of Leon	26	8.0
1387	b80344d063b5ccb3212f76538f3d9e43d87dca9e	Fireflies - Charttraxx Karaoke	24	9.0
1862	b80344d063b5ccb3212f76538f3d9e43d87dca9e	Horn Concerto No. 4 in E flat K495: II. Romanc...	23	10.0

```
In [26]: ir.get_similar_items(['Secrets - OneRepublic'])
no. of unique songs in the training set: 5151
Non zero values in cooccurence_matrix :639
```

Out[26]:

	user_id	song	score	rank
0		Use Somebody - Kings Of Leon	0.309524	1
1		Sehr kosmisch - Harmonia	0.303571	2
2		Dog Days Are Over (Radio Edit) - Florence + Th...	0.302326	3
3		Lucky (Album Version) - Jason Mraz & Colbie Ca...	0.263158	4
4		Drop The World - Lil Wayne / Eminem	0.257143	5
5		OMG - Usher featuring will.i.am	0.243243	6
6		Marry Me - Train	0.243243	7
7		Fireflies - Charttraxx Karaoke	0.238095	8
8		The Only Exception (Album Version) - Paramore	0.212121	9
9		Breakeven - The Script	0.200000	10

REFERENCE	1. Content-based filtering for music recommendation: A comprehensive survey by Bogdanov, D., Wack N., et al. (2013). 2. A survey on content-based filtering methods for efficient recommendation systems by Gupta, M., Mittal, N. (2016). 3. A content-based movie recommendation system using collaborative filtering and hybrid filtering techniques by Huang, J., Zeng, J., et al. (2016). 4. Content-based filtering using machine learning techniques for effective recommender systems" by Cambazoglu, B.B., et al. (2010).